From:	Simon Trimborn
To:	Inez Zwetsloot
Subject:	Fwd: Journal of Financial Econometrics - Decision on Manuscript ID JFEC-2026-014
Date:	maandag 4 mei 2026 22:39:51
Attachments:	Reviewer 1 - JFEC-2026-014.pdf

RR! Look at the comments, super positive!

I think we can be proud of ourselves. This is a top journal. An A on our side of the water and very highly regarded in the community. Let's chat this week and see how we want to address the points raised. I think we can make it through.

---------- Forwarded message ---------
Von: **Journal of Financial Econometrics** <onbehalfof@manuscriptcentral.com>
Date: Mo., 4. Mai 2026 um 21:54 Uhr
Subject: Journal of Financial Econometrics - Decision on Manuscript ID JFEC-2026-014
To: <trimborn.econometrics@gmail.com>

04-May-2026

Manuscript JFEC-2026-014: Reddit users unleashed - understanding user behaviour and their impact on meme stocks

Dear Prof. Trimborn,

Manuscript ID JFEC-2026-014 entitled "Reddit users unleashed - understanding user behaviour and their impact on meme stocks" which you submitted to the Journal of Financial Econometrics, has been reviewed. If not attached, the comments of the reviewer(s) are included at the bottom of this letter in your author center.

I have now received the reports of two referees and one associate editor. All the reviewers agreed that the paper studies an interesting topic and had high potential. They also made a list of issues that need to be addressed and recommended a major revision. The associate editor summarized well all the issues in his report, in particular the need to clarify the econometric contribution and the causal interpretation of the results.
Therefore, I invite you to revise your paper to address the various issues raised by the reviewers. When submitting your revision, you should also include a file with a detailed reply to all the comments of the reviewers.

When resubmitting your manuscript, please upload a pdf file, the latex files can be uploaded at a later stage for production purposes. Please check that your manuscript conforms to OUP guidelines, see instructions for authors on the website:

https://academic.oup.com/jfec

Please note all figures submitted to the journal in color will be published in color online at no cost (unless the author specifically requests that the figures be in black and white online). Figures will appear in black and white in the print issue. Color figures must have a resolution of at least 300 dots per inch at their final size. A figure legend that works for both color and grayscale should be submitted for each figure and provided at the end of the main document file.

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Or click on the following link:

*** PLEASE NOTE: This is a two-step process. After clicking on the link, you will be directed to a webpage to confirm. ***

https://mc.manuscriptcentral.com/jfec?URL_MASK=45ce9280f9b34432bb3878f00185f171

Please submit your revised manuscript before the due date: 29-Apr-2027.

If you require any assistance, or wish to request an extension, please do not hesitate to contact the Editorial Office at jfec.editorialoffice@theeditorialhub.com

Once again, thank you for submitting your manuscript to the Journal of Financial Econometrics and I look forward to receiving your revision.

Sincerely,

Marine Carrasco
Editor
Journal of Financial Econometrics

Referee(s)' Comments to Author:

Referee: 1

Comments to the Author
(There are no comments.)

Referee: 2

Comments to the Author
The paper at hand studies the relationship between individual user behaviour on the Reddit subreddit WallStreetBets and the market dynamics of the GameStop (GME) stock during the meme stock frenzy (and before and after). The authors propose an Interpretable Social Network and Market Behaviour modelling approach (ISMB), which combines logistic regression for link prediction at the node-pair level with LASSO and a change point detection algorithm. The approach identifies when individual users change their posting behaviour and aggregates these behavioural shifts into variables that are then linked to hourly and 5-minute returns, integrated volatility, and jump volatility of GME. The main findings are: (1) followers of influential users are more relevant for market dynamics than the influencers themselves, (2) influential users change their behaviour well before the market frenzy, and (3) the aggregated change point variables explain a meaningful share of market movements, outperforming standard network activity measures.
The paper makes a genuine contribution. The idea of modelling the social network at the individual user level, identifying structural breaks in posting behaviour, and linking those breaks to financial market outcomes is novel and well-motivated. The ISMB framework is methodologically interesting and the scale of the data collection (3.8 million posts, 502,650 users) is impressive.

However, the paper requires substantial revision along several dimensions, which I outline below.

Major Issues

1. The authors repeatedly position their work against studies that use aggregated network metrics. This is a fair motivation, because their individual-level modelling does reveal things that aggregate post counts do not (e.g., the timing of behavioural shifts across user types, the heterogeneity in parameter estimates). However, the market-impact analysis in Equation (7) uses CP_VN and CP_ANVN, which are themselves aggregates: counts of change points per hour across all nodes. So the final link between the network and the market is also an aggregated analysis. The authors do not look at individual trading behaviour.
In my reading, the advantage of the ISMB approach is not that it avoids aggregation, but that the aggregated variable is derived from individual-level modelling and therefore captures changes that a raw post count would miss. The framing should reflect this distinction more clearly rather than implying that the paper looks at the individual level throughout.

2. The paper repeatedly claims to study "changes in users' beliefs" following the DeGroot/Pedersen framework, but what the change point model actually captures is a structural shift in the parameters of a logistic link-prediction model on the communication probability between two nodes. This is not the same as a belief update. The mapping from estimated parameter shifts to belief changes is asserted but never really justified. The authors should be precise about what their measure captures (ideally already in the introduction). A change point means that the statistical relationship between the explanatory variables and the reaction probability has shifted. That is an interesting finding by itself, but calling it a change in beliefs overstates the interpretation. Related: The authors filter out the vast majority of users due to infrequent posting (only 28.1% posted four or more messages), which is a necessary step for estimation but means the analysis covers a selected subset of the network. Many users are lurkers who read but never post and may still trade based on what they read. This selection issue should be acknowledged and discussed.

3. The paper does not cite Allen et al. (2025). This paper studies the same January 2021 meme stock events, uses social media data from Reddit, Twitter, and StockTwits, and examines the link between social media activity, retail trading, and market outcomes. It provides detailed evidence on the securities lending market, options market activity, and market quality effects. The approaches are complementary: Allen et al. use aggregated mentions and a cross-sectional approach across 13 meme stocks, while this paper models the network at the individual level for a single stock. This comparison should be discussed. It would also strengthen the motivation for the individual-level approach, since it would clarify exactly what the ISMB framework adds relative to the aggregate analysis in Allen et al. (2025).

4. Regarding the Market impact regression, I have several issues.
(a) It is unclear to me from the table notes whether lagged return variables are included in all six model specifications or only in some. The note says "insignificant variables and significant lagged return variables are not displayed," which is ambiguous. The authors should confirm that all models include the same baseline controls. Maybe by adding an extra row?
(b) The coefficients are not interpreted economically. For example, VN3 in Table 5 equals -0.015, which means a change point of a vital node at lag 3 is associated with a negative return. This seems counterintuitive given the paper's narrative that VN behavioural change contributes to price increases, and it needs discussion. That the VN coefficients are much

larger in absolute terms than the ANVN coefficients (which are on the order of 0.0001) likely reflects the fact that the VN variable takes much smaller values (few change points per hour, as shown in Figure 7), whereas the ANVN variable aggregates over tens of thousands of users. Without standardization or a discussion of units and scaling, the reader cannot assess relative economic importance. This is important because the authors argue that VNs play a central role but that their behavioural change explains less of the market dynamics than ANVNs. The coefficients should be discussed in a way that makes this argument clear.
(d) As a robustness check, I would like to see the regressions re-estimated including only the variables that are statistically significant in the full model. With the large number of lags and variables, there is a risk that the high R-squared values are driven less by the statistical significant values but by just the entire set of explanatories.

5. The ISMB framework identifies when behaviour changes, but says very little about how it changes. Which parameters shift at the change point? Does the sensitivity to trading hours increase? Does the coefficient on the VN's activity become larger? The model produces this information but the paper largely ignores it. A discussion of what characterizes the behavioural shifts would substantially enrich the paper and provide more direct evidence for the Pedersen framework.
6. Lastly, the network activity (AC) is included as a feature in the link-prediction models (Equations 2 and 3). If the coefficient on AC is among the parameters that shift at the change point, then the change point variable may absorbs the information about network activity. This could explain why AC loses significance in the market regressions once CP variables are included, but it would not necessarily mean that individual behaviour change is fundamentally more informative than aggregate activity. The authors should address this.

Minor Issues
1. The paper needs significant editorial work.
(a) The introduction is repetitive. The main findings and the Pedersen/DeGroot framework are stated multiple times. A tighter introduction that motivates the gap, states the contribution once, and moves on would improve readability considerably.
(b) Section 2 ("Theory and Related Works") is largely repetitive to the intro. Either merge it into the intro or restructure it so that the intro provides the motivation and Section 2 delivers the formal theoretical backbone without overlap.
(c) The four stated contributions at the end of Section 1 overlap. Contributions I (empirical backing of Pedersen) and IV (quantifying the impact of social networks on financial markets) are essentially the same claim from different angles.
(d) I love the following sentence: "It is also remarkable that classical network measures [...] were unable to explain market movements as good as our designed change point variables". Regardless, I would tone it down.
(e) Consistency issues throughout: "U.S." vs. "US," references to "Appendix 6" when the appendix sections are lettered A through F, and other formatting issues.
(f) Several sentences are unnecessarily long and would benefit from splitting.

2. The source of the stock price data is not stated anywhere in the paper. Additionally, given that the GME frenzy was characterized by extreme trading volume, analyzing volume next to returns and volatility would be an interesting extension and would strengthen the paper.

3. Personally, I find the terminology "vital nodes" and "non-vital nodes" (and "admissible non-vital nodes") is a bit cumbersome. Terms like "influencers" and "followers" would be more intuitive and are already used informally in the paper and in the literature the authors

cite.

4. In Figure 2, the scatter plot (right panel) has a cluster of zero returns on weekends and holidays. Weekly returns might be a better choice for illustration purposes.

5. The reverse causality test is limited. The test only rules out a specific form of reverse causality (returns Granger-causing change points at hourly frequency). It does not address the possibility that both change points and returns are driven by common unobserved factors such as news events or specific posts by Keith Gill. The authors mention key events in Figure 8 but do not formally control for them. Including event dummies or discussing this limitation would strengthen the section. From a robustness and “causality” perspective, a placebo test with a completely unrelated stock would also make sense to show that the discussion/change points are really idiosyncratic to GME.

6. The change point detection parameters (window size of 2 months, step size of 1 week, minimum distance of 1 week between change points) are chosen somewhat arbirtrary. A sensitivity analysis (similar to the VN definition) varying these parameters would be useful.

7. The choice of the LASSO parameter and the cross-validation procedure are not discussed. In a time-series setting, standard iid cross-validation is inappropriate. The authors should clarify how cross-validation is implemented.

8. The 5-minute analysis (Section 5.3.2 and Appendix D/E) finds weaker results during the frenzy period and attributes this to external events (e.g., the Robinhood trading ban). This part needs a bit more substance or framing. For example, for the hour-by-hour the authors only find VN3 and 4 significant for the return. Those two are not even covered in the 5min regression (only 12 lags, thus 1h).

References
Allen et al. (2025): https://pubsonline.informs.org/doi/10.1287/mnsc.2023.02887

Associate Editor Comments to Author:

Associate Editor
Comments to the Author:
The paper has genuine promise. The combination of a very large Reddit network, interpretable link models, and change-point analysis is interesting, and the reports correctly note that the contribution could be substantial if it holds. That said, the novelty is not yet isolated cleanly enough relative to the existing dynamic-network and social-media/markets literatures. In particular, the market-link step ultimately uses aggregated counts of change points, so the paper's contribution is better described as constructing richer aggregate variables from individual-level models than as delivering a fully individual-level market-impact design. That distinction needs to be made much more clearly.

The identification is the main weakness. The paper repeatedly interprets breaks in the communication model as changes in users' beliefs, but that mapping is asserted more than identified. At present, the evidence is much more convincing as predictive or associational evidence than as causal validation of a belief-updating mechanism. The "vital node" classification is heuristic, even if some threshold robustness is provided, and the role of singular actors such as Keith Gill is not fully dealt with. The reverse-causality exercise is also limited; it does not do much to rule out common shocks, news, or event-driven confounding. That concern is reinforced by the weaker 5-minute return results during the key frenzy window, where the paper itself attributes patterns partly to outside events.

A related concern is selection. The empirical design excludes most low-activity users, since only 28.1% of nodes post four or more messages. That may be necessary computationally, but it limits the strength of claims about "the crowd," especially when lurkers may read and trade without posting. The paper should be more explicit about what population is actually being studied.

The econometric contribution relative to existing dynamic-network models is still blurry; the objective function, sparsity assumptions, cross-validation/LASSO choices, and sensitivity to change-point tuning need fuller and cleaner exposition; and the paper should say more about how behavior changes at the detected breaks, not just when those breaks occur. The reports also rightly note that the current draft needs tighter writing and more careful interpretation of coefficient magnitudes and scaling.

The paper is promising and the issues could be addressed in a revision. A revision would need to sharpen the novelty claim, substantially temper or better justify the causal/belief language, address the role of common shocks and major actors more directly, fully document estimation choices and data construction, and tighten the paper's presentation and positioning in the literature.

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